

DATA LINK LAYER (DLL)

OSI Model Layer 2 • Comprehensive Reference Guide

Definition:

- 2nd layer of the OSI Model.
- Responsible for node-to-node (device-to-device) communication.
- Converts raw bits from the Physical Layer into frames.
- Ensures reliable and error-free transmission.

Functions

- Framing
- Physical Addressing (MAC Address)
- Flow Control
- Error Detection
- Error Correction
- Error Control
- Medium Access Control (MAC)

Data Link Layer Issues

These are the major problems handled by the Data Link Layer.

1. Framing

- Divides data into frames.
- Adds header and trailer.

2. Physical Addressing

- Uses MAC addresses.
- Delivers data to the correct device.

3. Flow Control

- Prevents the sender from sending data too fast.
- Matches sender speed with receiver speed.

4. Error Control

- Detects and retransmits damaged or lost frames.

5. Error Detection

- Finds whether errors occurred during transmission.

6. Error Correction

- Corrects errors without retransmission (if possible).

Flow Control

1. Stop-and-Wait Protocol

Sender sends one frame at a time and waits for ACK before sending the next frame. If ACK is not received, it resends the frame.

- **Characteristics:** Very simple, low efficiency, suitable for small networks.
- **Sequence:** Frame → ACK → Frame → ACK
- **Advantages:** Simple, Easy to implement.
- **Disadvantages:** Slow, Wastes bandwidth.

2. Sliding Window Flow Control

Sender can send multiple frames before receiving ACK. Receiver stores frames in a window which moves (slides) after receiving ACK.

- **Advantages:** High efficiency, Better bandwidth utilization, Faster than Stop-and-Wait.
- **Disadvantages:** More complex.

Error Detection Algorithms

1. Parity Check

Adds one parity bit to detect single-bit errors. Types include **Even Parity** and **Odd Parity**.

- **Advantages:** Very simple, Low overhead.
- **Disadvantages:** Cannot detect all errors.

2. CRC (Cyclic Redundancy Check)

Uses binary polynomial division. Sender generates CRC bits; Receiver performs the same calculation. If remainder = 0, no error.

- **Advantages:** Highly accurate, Detects burst errors.
- **Disadvantages:** Slightly complex.

3. Checksum

Sender adds all data words and sends the checksum. Receiver repeats calculation. If checksum matches, data is correct.

- **Advantages:** Easy to implement, Widely used.
- **Disadvantages:** Less reliable than CRC.

Error Correction

Hamming Code

Detects and corrects single-bit errors using redundant (parity) bits. Receiver identifies the exact error position and corrects the bit without retransmission.

- **Advantages:** Automatic correction, No need to resend data.
- **Limitation:** Corrects only single-bit errors.

Error Control (ARQ)

ARQ = Automatic Repeat reQuest: If an error occurs, the receiver requests retransmission.

1. Stop-and-Wait ARQ

- Send one frame, wait for ACK. Timeout triggers resend.
- Features: Simple, Reliable, Slow.

2. Go-Back-N ARQ

- Sender sends multiple frames. If one is lost/error: Receiver discards that and all following frames. Sender retransmits from the erroneous frame onward.
- Advantages: Faster than Stop-and-Wait.
- Disadvantages: Retransmits many correct frames unnecessarily.

3. Selective Reject ARQ (Selective Repeat ARQ)

- Sender sends multiple frames. Receiver stores correct frames. Only the erroneous or lost frame is retransmitted.
- Advantages: Most efficient, Saves bandwidth.
- Disadvantages: More complex, Requires more memory.

HDLC (High-Level Data Link Control)

A bit-oriented DLL protocol developed by IBM and standardized by ISO. Supports point-to-point and multipoint communication.

Features: Bit-oriented, uses frames, CRC error detection, flow/error control, full/half-duplex, sliding window technique.

HDLC Frame Format

FLAG | ADDRESS | CONTROL | DATA | FCS (CRC) | FLAG

- **Flag:** Start/end of frame (01111110)
- **Address:** Receiver address
- **Control:** Frame type and flow/error control
- **Data:** Actual information
- **FCS:** CRC for error detection

Types of HDLC Frames

- **I-Frame (Information):** Carries user data.
- **S-Frame (Supervisory):** Flow and error control (ACK, NAK, etc.).
- **U-Frame (Unnumbered):** Link management and control.

Quick Comparison

Topic	Main Purpose
Stop-and-Wait Protocol	Flow control (1 frame at a time)
Sliding Window	Flow control (multiple frames)
Parity Check	Error detection (single-bit)
CRC	Error detection (burst errors, highly reliable)
Checksum	Error detection (addition method)
Hamming Code	Error correction (single-bit correction)
Stop-and-Wait ARQ	Error control (retransmit one frame)
Go-Back-N ARQ	Retransmit current and all following frames
Selective Reject ARQ	Retransmit only the erroneous frame
HDLC	Reliable bit-oriented Data Link Layer protocol

EXAM TIP (VERY IMPORTANT)

Remember the distinction:

- **Flow Control:** Controls transmission speed (Stop-and-Wait, Sliding Window).
- **Error Detection:** Finds errors (Parity Check, CRC, Checksum).
- **Error Correction:** Corrects errors (Hamming Code).
- **Error Control:** Handles errors using ACK/NAK and retransmission (Stop-and-Wait ARQ, Go-Back-N ARQ, Selective Reject ARQ).